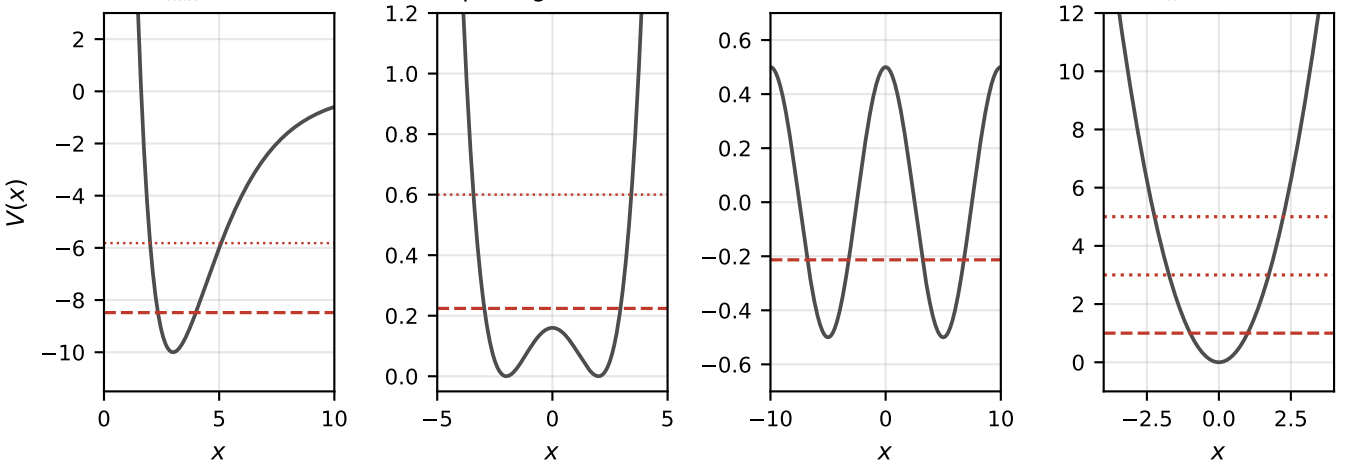
(a) Morse, $a = 0.5$, $x_0 = 3$
$V_{\min} = -10$
(b) double well
splitting $= 0.376$
(c) sinusoidal
$V \in [-0.5, 0.5]$
(d) harmonic
$E_n = 2n + 1$
$V(x)$
x